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# Resistor Combination Calculator
# Calculates total resistance in series
and in parallel.
# My twist: it also asks for a supply
voltage and works out the
# current and power for each circuit
using Ohm's law.


def get_positive_number(prompt,
whole_number=False):
    """Keep asking until the user types
a valid number greater than 0.
    If whole_number is True, the input
is converted with int(),
    otherwise it is converted with
float()."""
    while True:
        text = input(prompt)
        try:
            if whole_number:
                value = int(text)
            else:
                value = float(text)
        except ValueError:
            # The user typed letters or
symbols instead of a number
            print("Please type a
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number, not letters or symbols.")
            continue
        if value <= 0:
            print("The number must be
greater than 0.")
        else:
            return value


def series_resistance(values):
    """Series: just add all the
resistors. Returns the total in
ohms."""
    return sum(values)


def parallel_resistance(values):
    """Parallel: 1/Rt = 1/R1 + 1/R2 +
... Returns the total in ohms."""
    total_conductance = 0
    for r in values:
        total_conductance += 1 / r
    return 1 / total_conductance


def current_and_power(voltage,
resistance):
    """Ohm's law: I = V / R and P = V *
I. Returns current (A) and power
(W)."""
    current = voltage / resistance
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    power = voltage * current
    return current, power


def main():
    print("=== Resistor Combination
Calculator ===")

    while True:
        # Step 1: find out how many
resistors there are
        count =
get_positive_number("How many resistors
do you have? ", whole_number=True)

        # Step 2: store each resistor
value in a list
        resistors = []
        for i in range(1, count + 1):
            value =
get_positive_number(f"Enter value of
resistor {i} (in ohms): ")
            resistors.append(value)

        # Step 3: calculate the totals
        total_series =
series_resistance(resistors)
        total_parallel =
parallel_resistance(resistors)

        print("\n--- Results ---")
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        print(f"Resistors entered:
{resistors}")
        print(f"Series total:
{total_series:.2f} ohms")
        print(f"Parallel total:
{total_parallel:.2f} ohms")

        # Step 4 (my twist): current
and power from a supply voltage
        voltage =
get_positive_number("\nEnter the supply
voltage (in volts): ")
        i_series, p_series =
current_and_power(voltage,
total_series)
        i_parallel, p_parallel =
current_and_power(voltage,
total_parallel)

        print(f"\nAt {voltage} V:")
        print(f"Series:   current =
{i_series:.4f} A, power =
{p_series:.4f} W")
        print(f"Parallel: current =
{i_parallel:.4f} A, power =
{p_parallel:.4f} W")

        # Step 5: ask if the user wants
to go again
        again = input("\nCalculate
another circuit? (y/n):
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").strip().lower()
        if again == "y":
            print()
        elif again == "n":
            print("Goodbye!")
            break
        else:
            print("I did not understand
that, so I will stop here. Goodbye!")
            break


# This makes the program start when the
file is run
main()
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